

From Popularity to Prestige: Are Spotify Listeners Moving Closer to Grammy Standards? (2000–2023)

Abstract

For many years, people have wondered if the music that listeners enjoy is the same music that wins big awards. We tried to look at if Spotify users' preferences from 2000 to 2023 have gotten close to the songs that actually end up winning Grammy Awards. To achieve this, we brought together the Spotify Million Tracks dataset with a list of Grammy nominees and winners, matching songs by both track and artist name so we can find overlaps between the two datasets.

Our central visualization was an interactive histogram show-casing Spotify popularity by year. Over the years, Grammy-winning songs tended to have more and more popularity on Spotify. This suggests that listener preferred and Grammy-recognized music have been moving in the same direction.

Overall, our results show that Spotify users' tastes have become more similar to Grammy-winning music in recent years. There are still differences, but the gap between what people listen to and what the Grammys reward has gotten much smaller compared to twenty years ago.

1. Datasets

1.1 Data sources

This project uses two publicly available datasets:

- **Spotify 1 Million Tracks**

This dataset was taken from Kaggle via the URL: <https://www.kaggle.com/datasets/amitanshjoshi/spotify-1million-tracks>

It was originally collected from the Spotify platform using the Python library Spotipy. It contains about one million songs along with metadata and audio features. The file size is nearly 175 MB, with 20 columns for each track. It includes some main fields such as artist name, track name, track ID, genre, release year, and Spotify's popularity score. It also contains many audio features including valence, danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, tempo, and duration. Many of the metadata fields are strings, while popularity and year are integers, and all the audio features are floats. Hence, due to its large and detailed nature, it is perfect for studying music features and popularity patterns.

- **Grammy Award Nominees and Winners, 1958–2024**

This dataset was taken from Kaggle via the URL: <https://www.kaggle.com/datasets/kfoster150/grammy-award-nominees-and-winners-1958-2024>

It includes all Grammy nominations and winners, from 1958 to 2024, and comprises over 25,000 rows. Each line is a single nomination. The fields included are year and ceremony number, category of award, name of award given, work nominated, nominee, and whether the nomination won. In this dataset, most information is in string format, but 'Year' and 'Ceremony Number' are integers while the field indicating whether a nomination won is a

boolean. This data set enables views of professional recognition over the long term and can be combined with Spotify data to examine the correspondence between public taste and industry honors.

1.2 Big Data Characteristics

The datasets used in this project have many common big data features.

Volume: The Spotify dataset contains roughly a million tracks, providing a substantial sample for analysis.

Variety: The project combines two types of data: Spotify track features and Grammy award records. Since those kinds of datasets will differ in structure and content, they will require careful cleaning and matching before analysis.

2. Data Exploration, Processing, Cleaning and Integration

Before any visualizations could be created, both datasets required extensive preparation. We had to clean and standardize the names for the purpose to combine the Spotify and Grammy datasets as they come from two different sources and have different naming patterns.

2.1 Initial Exploration

We start by exploring each dataset separately to understand how they are structured:

- In the Spotify dataset, we examined the distribution of key audio features, verified the validity of the year and popularity fields, and checked for missing or obviously incorrect values such as negative durations.
- We have looked for inconsistencies in naming and duplicate entries in fields such as Award Name, Nominee, Work, and Year within the Grammy dataset.
- We also reviewed sample rows to understand how artists and track titles were formatted, since this would later impact how we match songs across datasets.

2.2 Cleaning the Spotify Dataset

For cleaning the Spotify data, a number of steps were taken:

- Removed rows with missing or invalid popularity values.
- Standardized text fields - track_name and artist_name - by converting to lowercase and trimming whitespace.
- Filtered the dataset to only include songs released between 2000 and 2023.
- Converted audio feature fields to consistent numeric formats.
- Removed obviously incorrect values, such as tempo = 0 and very short durations.

After these steps, the dataset was more reliable and ready for matching.

2.3 Cleaning the Grammy Dataset

For the Grammy dataset:

- We standardized song titles and artist names using similar methods as for the Spotify dataset: lowercasing and removing punctuation.
- Removed duplicate nominations for the same song, where necessary.

2.4 Matching Songs Across Datasets

Once both datasets were cleaned, we tried matching songs on track title + artist name. Since the two datasets weren't originally designed to link together, we had to account for small variations like:

- "feat." vs. "featuring"
- "&" vs. "and"
- punctuation differences
- alternate versions of songs (live, remix, etc.)

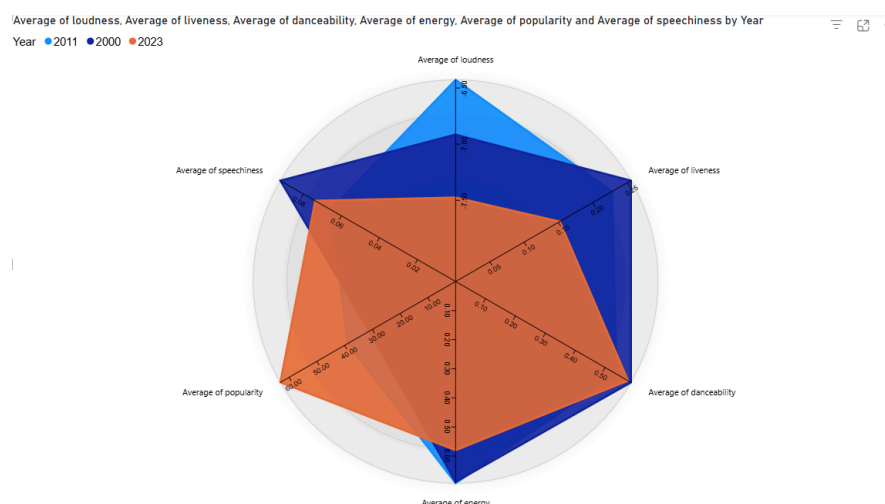
To handle such inconsistencies, we applied string-matching rules. After matching, we extracted the subset of songs appearing in both datasets. This merged dataset represents the songs which are:

- present in Spotify's large library,
- recognized by the Grammys,
- and fall within the years 2000–2023.

The final integrated table served as the base for all the visualizations and analyses in this project.

3. Visualisation

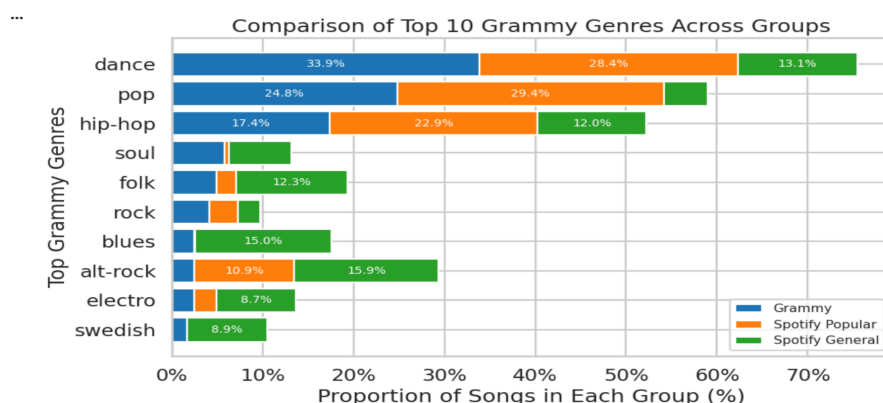
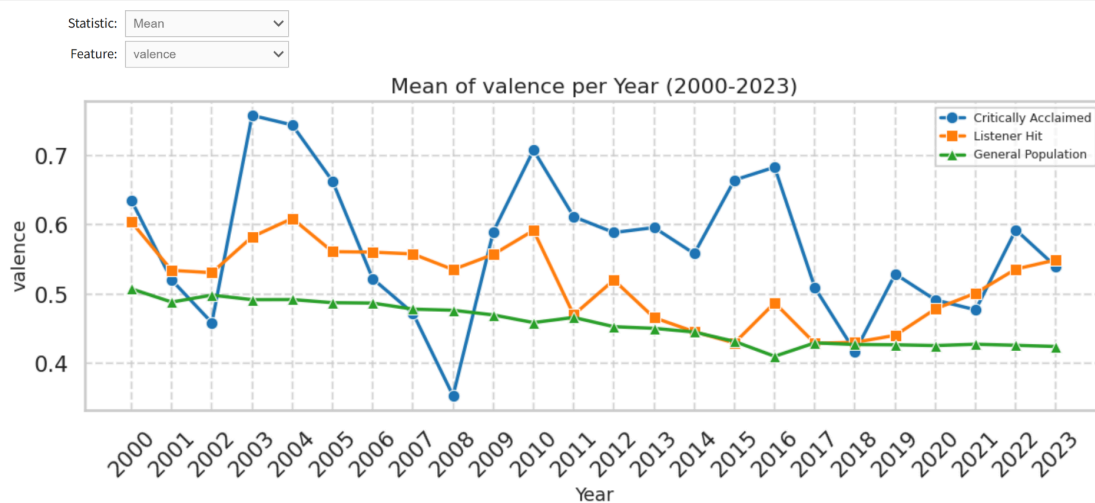
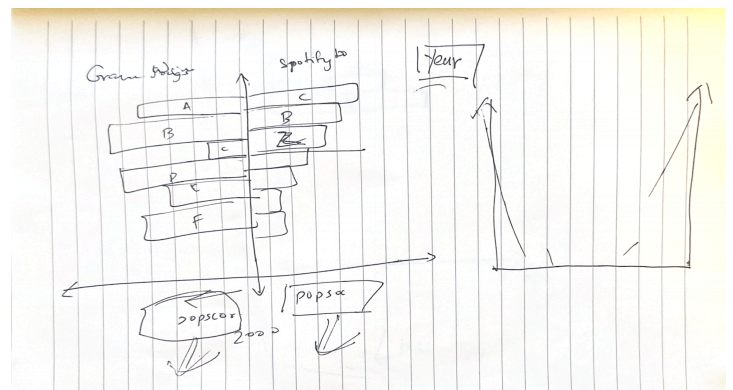
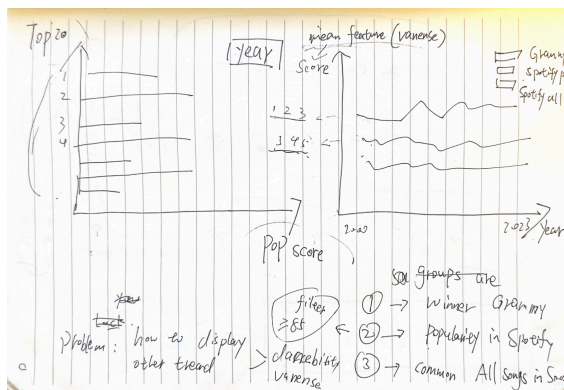
3.1 Design Process



The design process started by broadly brainstorming to find the best way to show changes over time and differences between Spotify popularity and recognition from the Grammy Awards. The team first sketched a number of ideas for showing trends over the years with line graphs, comparing two variables at once with dual-axis charts, and comparing multiple

audio features with radar charts. These early sketches helped guide discussions, but each had clear drawbacks for answering the main research question.

This involved, for example, separating the songs into groups such as Grammy recognized tracks versus different sub-sets of Spotify songs and plotting the average popularity or valence for each of those groups year to year. But this meant reducing all of the data into single averages that obscured important differences within each group. Additionally, some Grammy categories had very few matched songs (only looking at "Record of the Year" nominations, for example, left too little data to form reliable trend lines). Because of this, the line-chart approach couldn't clearly show how the full popularity distribution was changing over time.



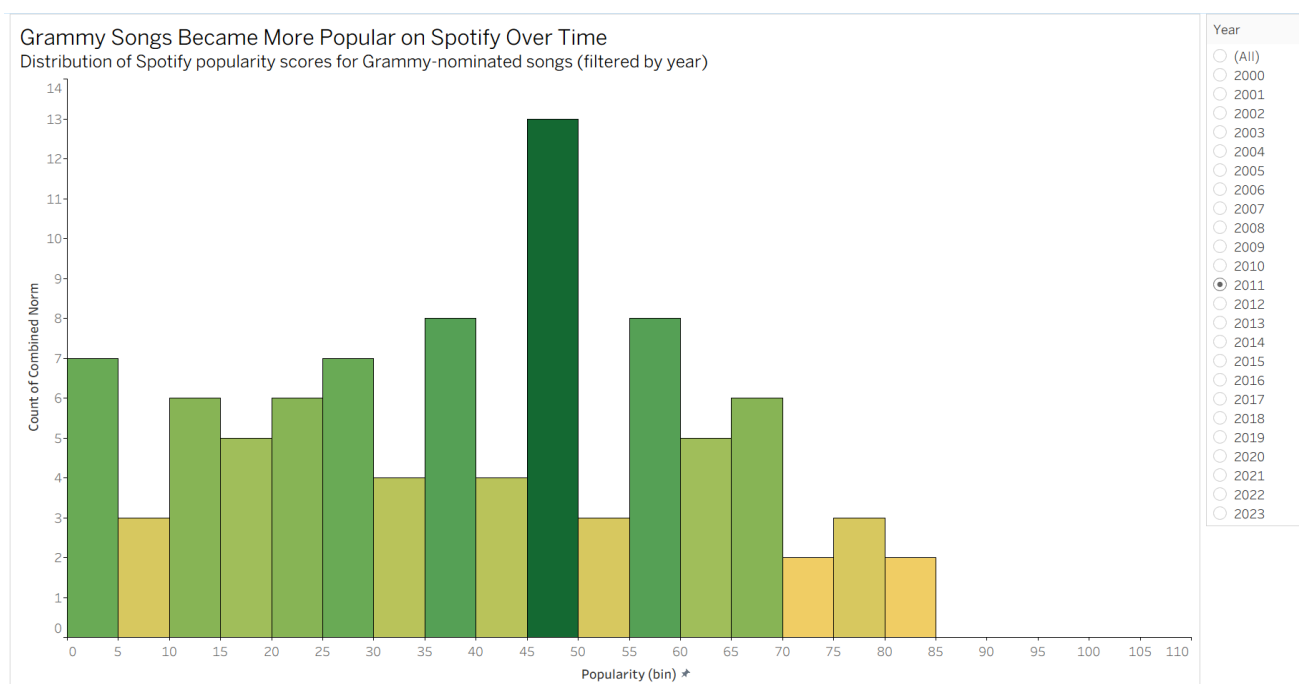
Other chart types were tested: the team drew bar charts to display the number or percentage of songs per year, but these either eliminated the continuous time aspect or were messy after being stretched across many years. Dual-axis radar charts were considered for displaying multiple audio features at once, but they are hard to read and not ideal for tracking change over time. For this, dual-axis line charts risk confusing the viewer, given the two scales combined on one plot. These tests made one thing quite clear: the visualization needed to highlight how the entire distribution of popularity changed from year to year, without oversimplifying or adding unnecessary complexity.

With that in mind, the team moved toward distribution-based visualizations. One of the main options was an interactive histogram of Spotify popularity. The chart shows how many songs fall into each range of popularity, and it has a filter on year for users to select one year. Thus, viewers may easily see, by examining one year at a time, how the entire popularity distribution of Grammy-recognized songs shifts over time. That is a direct answer to the central question: whether Grammy-recognized songs have moved into higher ranges of popularity, meaning that public and professional tastes are becoming more aligned.

Histograms provide a way to visualise the distribution of data. For example, how spread out the data is, if any skew occurs, and how the data distribution evolves through the various years of the time period analysed. Therefore, it was determined that the best way to present the analysis of data would be to create an interactive histogram where a timeline filter allows users to see how the data has changed over time.

Each stage of the design process has contributed to selecting the interactive histogram over the other three options. The histogram allows for easy identification of trends over the years; notably, Grammy-winning songs have been gaining popularity on Spotify at an increasing rate over the years, illustrating that as time passes, the gap between the type of music people listen to and the type of music that wins awards continues to diminish.

The following is a screenshot of our final visualization:



3.2 Why a histogram?

A histogram works well in this project, since it can really reflect how the overall spread of popularity changes over time. Instead of looking at individual songs, it shows the greater picture-whether the distribution shifts, becomes more skewed, or moves toward higher or lower popularity ranges.

Our central question is whether the public listening habits are becoming more similar to the songs recognized by Grammy Awards. Since our question is comparing two popularity distributions and seeing how close or far apart they are, a histogram is the easiest and most effective way to show that.

3.3 Design choices

The final histogram aimed to clearly present the pattern, most importantly how the popularity of Grammy-nominated songs changes over time. Every design choice in terms of color, layout, labels, and interactivity has been done with that goal in mind. Many of our decisions were informed by suggestions on [Evergreen & Emery's Data Visualization Checklist \(2016\)](#).

Use of Colour

A green-to-yellow color scale was chosen so that viewers can get a sense of where most songs fall within the range of popularity. Darker greens call out areas of high density for Grammy-nominated songs, while light yellows represent areas of low density. The palette is very simple and easy to view on a white background, and avoids color combinations which cause accessibility issues. Because it utilizes one continuous color scale, the viewer's attention remains on the density rather than on multiple competing colors.

Marks and Shapes

A histogram was chosen since it gives the clearest view of a continuous variable such as Spotify popularity. All the bars are plain and simple, in 2D, without shadows or 3D effects; that way, the viewer's attention goes directly to the pattern of variation in the data rather than to decorative elements.

Layout and Structure

The title clearly gives the overall message ("Grammy Songs Became More Popular on Spotify Over Time"), so the readers immediately know the purpose of the chart. A small subtitle puts the view in context. The labels for the axis are simple and easy to read, while the grid lines are light, so they support readability without distracting from the histogram. The whole layout is neat; the visual weight of the bars is predominant, while the supporting elements remain subtle.

Typography and Labels

There is a clear hierarchy here: the title is most noticeable, then the subtitle, followed by the axis labels and finally the tick labels. We don't put numbers on each bar since that would cause clutter, the axis already conveys the scale needed. We fixed the scales so the pattern can be analysed with ease when changing years.

Interactivity

This is a very important interactive feature: the year filter. If all years were to be shown simultaneously, it would make the chart too busy; if one selects a given year, then this allows users to explore how the distribution would change from 2000 to 2023. That gives support to the key question of whether the popularity of Grammy-recognized songs is gradually moving closer to public preference.

Overall Cohesion

Each design decision comes together to reinforce the key story through the color scale, clean layout, limited labeling, and year filter. The final visualization shows in a simple, clear, and focused way how the popularity of Grammy-nominated songs has changed over time.

4. Conclusion

List of tools: For this project, we mainly worked with Python in Jupyter Notebook using libraries like pandas, matplotlib, and seaborn to clean the data, preprocess it, and run exploratory analysis. The final explanatory histogram was built in Tableau. During our early exploratory stage, we also used Power BI to create radar charts and clustered column charts. In terms of teamwork, file sharing was managed via Google Drive to ensure that everyone was able to edit and update the files without a hitch.

Outcome: Our goal is to know if Spotify user's tastes of music are becoming more aligned with the Grammy Award's music standards. We have made histograms comparing the popularity of Grammy winners' songs each year; clearly showing the trend is towards Grammy winners being much more popular than they were at the beginning of the 2000s. The trend from many low-popularity Grammy winners during the early years of the 2000s to many high-popularity Grammy winners by 2023, indicates that professional award selections are becoming closer in alignment with what the public is enjoying these days.

Limitations and Future Improvements: There were a number of challenges in combining the two datasets. We used the combination of artist name and song title to match songs, which was made very difficult due to naming differences across platforms. For instance, variations such as "Doja Cat Featuring SZA" vs. "Kiss Me More (feat. SZA)" or "Lady Antebellum" vs. "Lady A" created mismatches. Even after cleaning, splitting, and standardizing names, some song titles were not matched correctly. This process can be more on point if it uses fuzzy-matching tools or more advanced alignment tools in the future.

We could enhance the analysis further by adding other audio features such as energy, acousticness, or loudness to compare how different musical qualities shift across time. More dynamic visual features in Tableau, such as automatic animation by year, would also help depict the narrative more clearly.

Team Collaboration: Shubin Li and Ananya worked equally to complete the project. Both team members provided contributions to data cleaning, exploratory analysis, and visualization, and their ideas influenced the project's ultimate course.

At the start, the team had different ideas for how to approach the visualization. Shubin Li suggested dividing the dataset into three groups; the Grammy group, the Spotify-Pop group, and the general Spotify reference group, and plotting time-series line charts. His approach included using dual filters so users could switch between different audio features on the

Y-axis, such as valence, danceability, energy, and instrumentalness, as well as switching between statistical methods like mean and median. He also developed several exploratory charts, including the valence trend line chart, a genre comparison chart, and a category-ranking visualization. These visuals helped the team understand broad patterns in how listener behaviour and award trends changed over time in relation to the Grammy Awards.

Ananya preferred to compare the yearly performance of Grammy-winning tracks directly with the most popular songs on Spotify. Her idea focused on using a year filter to examine how much overlap appeared between the two groups each year. She carried out major parts of the data cleaning and standardization, built the radar chart in Power BI, and created the final interactive histogram in Tableau. Her data pipeline after cleaning, merging, and aligning the datasets offered more complete year-wise coverage, which made it suitable as the foundation for the final explanatory visualization.

Both team members worked on cleaning and matching the datasets, especially where naming differences (such as artist variations) made alignment difficult. Although Shubin Li's focus on the "Record of the Year" category produced valuable insights, the smaller number of matches in that category made long-term patterns harder to represent. As a result, the project used Ananya's merged dataset for the final visualization, while Shubin Li's exploratory work played a key role in shaping the analytical direction and showing which features were most meaningful to explore.

Throughout the project, Ananya and Shubin Li shared intermediate outputs and updated files through Google Drive, which kept the workflow organized and transparent. Their combined efforts resulted in a well-rounded understanding of how Grammy recognition and public listening preferences have changed over time, while also giving both team members strong experience in data cleaning, cross-source alignment, and building clear visual narratives.

References

[Kaggle: Spotify 1 Million Tracks](#)

[Kaggle: Grammy Award Nominees and Winners, 1958–2024](#)

[Tableau Tutorial](#)

[Evergreen's Data Visualization Checklist](#)

We used AI at the initial stage to explore problem statements we could potentially work on.